THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For HCF(18, 24), Student A gets 6 and Student B gets 72. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

2. The HCF of 8, 9 and 25 is: 1

(a) 8 **(b)** 9 **(c)** 25 **(d)** 1

3. Which is a zero of $x^2 - 3x + 2$? 1

(a) -2 **(b)** -1 **(c)** 2 **(d)** 3

4. The zeroes of $x^2 + 2x - 15$ are: 1

(a) $3, -5$ **(b)** $15, -1$ **(c)** $5, -3$ **(d)** $-3, 5$

5. A student states: "If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, the lines are parallel." The correct response is: 1

(a) Correct **(b)** They coincide

(c) Incorrect — the lines intersect **(d)** Cannot be determined

6. The lines $x + y = 4$ and $x - y = 0$ intersect at: 1

(a) $(0, 4)$ **(b)** $(1, 3)$ **(c)** $(4, 0)$ **(d)** $(2, 2)$

7. A student states: "If $D = 0$, the quadratic equation has no roots." The correct response is: 1

(a) Correct **(b)** It has three roots

(c) Incorrect — it has two equal real roots **(d)** Cannot be determined

8. Which equation has equal roots? 1

(a) $x^2 - 8x + 16 = 0$ **(b)** $x^2 + 16 = 0$ **(c)** $x^2 - 16 = 0$ **(d)** $x^2 - 8x + 15 = 0$

9. A student states: "$1, 4, 9, 16, \ldots$ is an AP because the numbers increase." The correct response is: 1

(a) Correct

(b) Incorrect — the differences 3, 5, 7 are not equal

(c) Correct with $d = 3$

(d) Cannot be determined

10. The sum of the first 100 natural numbers is: 1

(a) 4950 **(b)** 10100 **(c)** 5000 **(d)** 5050

11. $\triangle PQR \sim \triangle XYZ$ with $PQ = 3$ cm, $XY = 6$ cm and $YZ = 10$ cm. Then QR is: 1

(a) 4 cm **(b)** 5 cm **(c)** 6 cm **(d)** 20 cm

12. Two triangles with all corresponding angles equal are similar by the criterion: 1

(a) AAA **(b)** RHS **(c)** SSS **(d)** SAS

13. The value of $\cos^2 30^\circ - \sin^2 30^\circ$ is: 1

(a) 0 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** $\frac{\sqrt{3}}{2}$

14. If OP is a radius and PT the tangent at P, then $\angle OPT$ is: 1

(a) 60° **(b)** 180° **(c)** 45° **(d)** 90°

15. The arc length of a quadrant of radius 28 cm is: 1

(a) 176 cm **(b)** 44 cm **(c)** 88 cm **(d)** 22 cm

16. If the area of a circle is 616 cm², its radius is: 1

(a) 7 cm **(b)** 28 cm **(c)** 14 cm **(d)** 21 cm

17. The curved surface area of a cone of radius 6 cm and height 8 cm is: 1

(a) 96π cm² **(b)** 60π cm² **(c)** 36π cm² **(d)** 48π cm²

18. If the mean is 30 and the mode is 24, the median is: 1

(a) 27 **(b)** 28 **(c)** 26 **(d)** 32

19. The mean of 2, 4, 6, 8, 10 is: 1

(a) 5 **(b)** 30 **(c)** 6 **(d)** 8

20. The probability of a number greater than 6 on a die is: 1

(a) 1 **(b)** $\frac{1}{6}$ **(c)** $\frac{5}{6}$ **(d)** 0

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Classify as rational or irrational with reason: 2

(i) $(3 + \sqrt{7}) - \sqrt{7}$

(ii) $(\sqrt{2} + \sqrt{3})^2$.

22. If α, β are the zeroes of $3x^2 + 5x - 2$, find $\alpha^2 + \beta^2$. 2

23. Solve by factorisation: $6x^2 - x - 2 = 0$. 2

24. In $\triangle ABC$, $DE \parallel BC$ with $AD = 4x - 3$, $AE = 8x - 7$, $BD = 3x - 1$ and $CE = 5x - 3$. Find x. 2

25. Find the coordinates of P on AB such that $\frac{AP}{PB} = \frac{2}{3}$, where A(−1, 3) and B(9, 8). 2

26. Evaluate: $4\cot^2 45^\circ - \sec^2 60^\circ + \sin^2 60^\circ + \cos^2 90^\circ$. 2

27. From the top of a 20 m building, the angle of depression of a car is 30°. Find the distance of the car from the building. 2

28. The tangent from a point 25 cm from the centre of a circle is 24 cm long. Find the radius. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $2x + 3y = 12$ and $x - y = 1$. 3

30. Find the mean by the step-deviation method: 3

Class	0–50	50–100	100–150	150–200	200–250	250–300
Frequency	4	8	16	13	6	3

31. If the 9th term of an AP is zero, prove that its 29th term is twice its 19th term. 3

32. Prove that the tangents drawn from an external point to a circle subtend equal angles at the centre. 3

33. A pie-chart circle of radius 7 cm has a sector showing 30% of the total. Find its angle and area. 3

34. A carton of 50 eggs has 5 rotten eggs. 3
(i) Find the probability that an egg drawn is good.
(ii) If 3 good eggs are first removed, find the probability that the next egg is rotten.

35. A kite is flying at a height of 75 m. Its string makes 60° with the ground. Find the length of the string. 3

OR

A man on top of a 100 m cliff sees two boats in the same direction at angles of depression 60° and 30°. Find the distance between the boats. 3

36. Find the median of: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	2	12	22	8	6

OR

Find the mode of: 3

Class	1–4	4–7	7–10	10–13	13–16	16–19
Frequency	6	30	40	16	4	4

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Show that A(−1, 0), B(3, 1), C(2, 2) and D(−2, 1) are the vertices of a parallelogram. Is it a rectangle? Justify. 5

OR

The vertices of $\triangle ABC$ are A(−1, 3), B(1, −1) and C(5, 1). 5
(i) Find the length of the median through A.
(ii) Show that the triangle is right-angled and isosceles.

38. Prove that $\frac{1+\cos A}{\sin A} + \frac{\sin A}{1+\cos A} = 2\csc A$. Verify it for $A = 60^\circ$. 5

OR

Prove that $(\sin A + \cos A)^2 = 1 + 2\sin A\cos A$. Hence, if $\sin A + \cos A = \sqrt{2}$, find $\sin A \cos A$ and A. 5

39. A solid is a cylinder of radius 7 cm and height 10 cm, with a cone of height 24 cm on top and a hemisphere at the bottom (same radius). Find 5
(i) its volume
(ii) its total surface area.

OR

Case study: A trophy has a cylindrical base of radius 5 cm and height 4 cm with a hemisphere of radius 5 cm on top. (Use $\pi = 3.14$) 5

(i) Find its volume.

(ii) Find the curved area to be gold-plated.

(iii) Find the cost of plating at ₹2 per cm².

40. The mean of the following distribution is 57.6 and $\Sigma f = 50$. Find f_1 and f_2. 5

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	7	f_1	12	f_2	8	5

OR

Find the mean, median and mode of the daily wages: 5

Wages (₹)	200–250	250–300	300–350	350–400	400–450
Workers	5	12	18	10	5

END OF QUESTION PAPER **Total: 80 Marks**

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